

EXPERIMENT 2: DDL Commands with Constraints – PRIMARY, FOREIGN KEY, UNIQUE, CHECK

```
mysql> CREATE TABLE STUDENT2 (  
->     RegNo INT NOT NULL,  
->     S1Name VARCHAR(50),  
->     PRIMARY KEY (RegNo)  
-> );  
ERROR 1050 (42S01): Table 'student2' already exists  
mysql> DESC STUDENT2;  
+-----+-----+-----+-----+-----+-----+  
| Field | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| RegNo | int           | NO   | PRI | NULL    |       |  
| S1Name | varchar(50)   | YES  | UNI | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
2 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE FACULTY_C (  
->     FacNo VARCHAR(4) PRIMARY KEY,  
->     Gender CHAR(1) CHECK (Gender IN ('M', 'F'))  
-> );  
Query OK, 0 rows affected (0.04 sec)  
  
mysql> DESC FACULTY;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| FacNo      | varchar(4)    | YES  |     | NULL    |       |  
| FacName    | varchar(15)   | YES  |     | NULL    |       |  
| Gender     | char(1)       | YES  |     | NULL    |       |  
| DOB        | date          | YES  |     | NULL    |       |  
| DOJ        | date          | YES  |     | NULL    |       |  
| MobileNo   | bigint        | YES  |     | NULL    |       |  
| Dept       | varchar(10)   | YES  |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
7 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE DEPARTMENT_C (
->     DeptNo VARCHAR(4) PRIMARY KEY,
->     DeptName VARCHAR(15),
->     DeptHead VARCHAR(4)
-> );
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> DESC DEPARTMENT;
```

Field	Type	Null	Key	Default	Extra
DeptID	int	NO	PRI	NULL	
DeptName	varchar(50)	YES		NULL	

2 rows in set (0.00 sec)

```
mysql> CREATE TABLE COURSE_C (
->     CourseNo VARCHAR(3) PRIMARY KEY,
->     CourseDesc VARCHAR(14),
->     CourseType CHAR(1),
->     SemNo CHAR(1) CHECK (SemNo IN ('1', '2', '3', '4', '5', '6')),
->     HallNo VARCHAR(4),
->     FacNo VARCHAR(4)
-> );
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> DESC COURSE;
```

Field	Type	Null	Key	Default	Extra
CourseNo	varchar(3)	YES		NULL	
CourseDesc	varchar(14)	YES		NULL	
CourseType	char(1)	YES		NULL	
SemNo	char(1)	YES		NULL	
HallNo	varchar(4)	YES		NULL	
FacNo	varchar(4)	YES		NULL	

6 rows in set (0.00 sec)

```

mysql> CREATE TABLE STUDENT2_UQ (
->     RegNo INT PRIMARY KEY,
->     S1Name VARCHAR(50) UNIQUE
-> );
Query OK, 0 rows affected (0.05 sec)

mysql> DESC STUDENT2;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int           | NO   | PRI | NULL    |       |
| S1Name | varchar(50)   | YES  | UNI | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> CREATE TABLE STUDENT3_CHILD (
->     RegNo INT,
->     S2Name VARCHAR(50),
->     FOREIGN KEY (RegNo) REFERENCES STUDENT2_PARENT(RegNo)
-> );
ERROR 1824 (HY000): Failed to open the referenced table 'student2_parent'
mysql> SHOW TABLES;
+-----+
| Tables_in_college |
+-----+
| course             |
| course_c           |
| department         |
| department_c       |
| faculty            |
| faculty_c          |
| student            |
| student2           |
| student2_uq        |
+-----+
9 rows in set (0.00 sec)

```